SHAGUFTA ANJUM

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EDUCATION

University of Illinois Urbana-Champaign Illinois, USA | Jan 2024 - April 2025

Master of Computer Science *GPA: 4/4*

Courses: Cloud Computing, Distributed Systems, Applied Machine Learning, Data Mining, Large Language Models, Computer Vision, Human Computer Interaction

Mahindra University India | July 2019 - May 2023

Bachelor of Technology, Computer Science Engineering *GPA: 3.95/4*

EXPERIENCE

Cognida.ai Sunnyvale, CA, USA

Software Engineer – Applied AI *Jan 2026 – present*

- Built an AI-assisted entity-mapping system using LLM inference to automate financial data migration across heterogenous ERP systems
- Implemented confidence scoring, evaluation metrics and human-in-the-loop validation workflows to improve entity matching accuracy
- Designed backend services to ingest and normalize ERP data from both APIs and CSV exports for downstream AI processing pipelines
- Developed embedding-based similarity search to retrieve candidate matches and orchestrated LLM prompts to boost response accuracy

Amazon New York, NY, USA

Software Engineer *June 2025 – Dec 2025*

- Designed and implemented a critical Bulk User Management feature for Amazon Ads to replace manual workflows with streamlined operations
- Built distributed scalable Spring Boot APIs supporting large-scale user operations with hierarchical permission checks across 10K+ accounts
- Architected an algorithm for account-hierarchy nesting limits using Redis-cached relationship graphs, reducing authorization latency ~35% at scale
- Set up AWS AppConfig integration for feature gating and runtime configurations, enabling safer rollouts and controlled experimentation
- Parallelized backend workflows for large batch operations, reducing processing time ~40–50% and improving system throughput under peak load
- Partnered with Product/QA during UAT and launch; identified and resolved critical defects and coordinated cross-team access-control changes

GlobalLogic Inc. Santa Clara, CA, USA

Software Engineer Intern *June 2024 – July 2024*

- Designed an LLM-powered assistant using OpenAI API integration, LangChain & Pinecone to answer user questions on a learning platform
- Architected a retrieval augmented generation (RAG) pipeline using semantic embeddings and vector search to dynamically retrieve relevant context
- Built a scalable FastAPI backend for real-time inference, implementing async request handling to support low-latency query execution
- Containerized the system with Docker and Kubernetes to enable reliable deployment and scaling to hundreds of concurrent users

Dell Technologies Bengaluru, India

Software Engineer *July 2023 – Dec 2023*

- Developed a computer vision pipeline to automate defect detection in packaging lines using images captured from edge devices
- Fine-tuned CNN models for text localization and blur detection using PyTorch and integrated with AWS Sagemaker for hosting and inference
- Optimized data ingestion and streaming of the image metadata and inference outputs using an event-driven Kafka pipeline
- Deployed containerized Python backend services on AWS to invoke model inference and publish prediction outputs to Kafka topics

PROJECTS

Personalized Writing Assistant using Fine-tuned LLMs *Sep 2024 – Dec 2024*

- Fine-tuned open-source LLMs with LoRA modules on domain-specific writing data to support personalized enterprise-style content generation
- Built a retrieval augmented generation (RAG) pipeline for inference using semantic embeddings for contextual retrieval and generation
- Implemented prompt orchestration, guardrails and an evaluation pipeline and designed backend API inference APIs for real-time generation

VisionAid – Smart Assistance for the Visual Impaired *June 2023 – Nov 2023*

- Designed a computer vision system to assist blind individuals in locating household items with real-time object detection using vision models
- Trained the models on a custom dataset with image augmentations, and designed a voice-enabled web app to provide audio-based navigation cues
- Optimized inference latency and model selection for real-world usability on constrained devices

SKILLS

- Programming Languages: Python, Java, JavaScript, Typescript, SQL
- Web Development: React.js, Django, Angular, Node.js, Spring Boot, Flask, FastAPI
- Databases: MongoDB, MySQL, PostgreSQL, Redis, Pinecone
- Machine learning: PyTorch, TensorFlow, Scikit-learn, Pandas, Huggingface
- Cloud and DevOps: AWS, GCP, Docker, Kubernetes, CI/CD, GitHub Actions, Kafka